

Runtime Operations of a Tempus-Based Partition Trigger: Kernel Instantiation, Bisimulation Proof, and the Complete Closed Loop from Partition Physics to Dispatch

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Abstract

This paper closes the loop between the partition-physics derivations of this monograph series and the Tempus/Buhera dispatch substrate: we prove that a well-typed Tempus trigger program, compiled to vaHera and dispatched by the five-subsystem kernel, computes the same partition classification as the direct partition-physics derivation on every input event (Theorem 4.4 — the Tempus-Physics Bisimulation).

We instantiate the five kernel subsystems $\mathcal{K} = \langle \text{Exec}, \text{PVE}, \text{CMM}, \text{PSS}, \text{DIC}, \text{TEM} \rangle$ explicitly for the LHC trigger: PVE is the partition-coordinate refinement type checker (rejecting selection-rule-violating fragments as type errors at static time); CMM is a balanced timing-interval tree keyed on $\Delta P(k)$; PSS is the parallel dispatcher for the four commuting detector subsystems; DIC is the HLT retrieval controller; TEM is the online composition-efficiency monitor.

We prove six operational theorems: (i) the 25 ns bunch crossing window is computed from detector geometry via the universal transport primitive $\tau_{\text{BX}} = \Delta x_{\text{ATLAS}}/c$ (Theorem 3.1); (ii) parallel PSS dispatch is correct because the four detector subsystems have diagonal coupling in the universal transport formula — the runtime realisation of the QND commutation theorem (Theorem 2.2); (iii) the timing-only trigger has structural incorruptibility because the PVE’s frozen surface accepts only $\Delta P(k)$ -typed fragments, giving zero injection surface (Theorem 3.6); (iv) the composition efficiency η_C is the normalised throughput coefficient X/Σ of the universal transport primitive evaluated on the trigger menu (Theorem 3.3); (v) three-route mass verification runs as a single PVE conformance test at $\mathcal{O}(1)$ per dispatch at LHC Planck depth (Theorem 3.5); (vi) the kernel’s Dispatch Correctness and Stability Contract theorems guarantee that adding HL-LHC trigger paths does not invalidate any existing client of the frozen interface (Theorem 3.7).

We trace a complete end-to-end di-muon trigger execution (Section 5), walking a single event from raw

detector signals through Tempus compilation, vaHera dispatch, type checking, backward trajectory completion, parallel subsystem evaluation, and L1 accept/reject decision, with every step formally grounded in a theorem of the partition algebra or the dispatch substrate.

Keywords: Tempus trigger; Buhera kernel; vaHera dispatch; partition physics; bisimulation; timing window; parallel dispatch; structural incorruptibility; composition efficiency; di-muon trigger; closed loop

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1 Introduction

1.1 The closure problem

The physics papers of this monograph series derive, from first principles, every design parameter of the LHC trigger: the 25 ns timing window, the Planck depth $n_P = 60$, the four-subsystem CSCO, the backward trajectory complexity $\mathcal{O}(\log_3 N)$, the selection rules as structural dead zones, and the composition efficiency asymptote. The operating-system papers derive, from three axioms about bounded categorical processors, a complete dispatch substrate: refinement types, three-route verification, universal transport, operational bisimulation, and a five-subsystem kernel.

This paper proves that the two frameworks describe the same object. The LHC trigger IS the dispatch kernel instantiated with partition-physics parameters. The Tempus trigger program IS a vaHera fragment over partition-coordinate refinement types. The trigger's L1 decision IS the kernel's dispatch output. These are not analogies — they are identities provable by operational bisimulation.

1.2 The Tempus language and its compilation to vaHera

Definition 1.1 (Tempus Trigger Program). A *Tempus trigger program* P is a finite sequence of *cell definitions* and *firing rules*:

$$C_i = \{k \mid |\Delta P(k)| < \delta_i\} \quad (\text{timing cell}) \quad (1)$$

$$P = \bigvee_{i \in I} \left(\bigwedge_{k \in K_i} C_i(k) \right) \quad (\text{trigger condition}) \quad (2)$$

where $\Delta P(k) = T_{\text{ref}}(k) - t_{\text{rec}}(k)$ is the timing residual of channel k , $\delta_i \in \mathbb{R}^+$ is the cell half-width, K_i is a required set of channels, and the disjunction is over trigger paths $i \in I$.

Definition 1.2 (Compilation to vaHera). The *compilation* $\llbracket P \rrbracket$ of a Tempus trigger program to a vaHera AST proceeds as follows:

- (i) Each *detector hit* $h_k = (k, t_{\text{rec}}(k))$ is a $\text{Literal}(h_k)$, a literal carrying the raw timing measurement.
- (ii) Each *detector subsystem* (tracker, ECAL, HCAL, muon) is a $\text{Call}(\text{Op}_s, h_k)$, a call to the refined operation Op_s that extracts partition coordinate $s \in \{n, \ell, m, s\}$.
- (iii) Each *trigger path* i is a $\text{Compose}(\text{Call}(\text{Op}_n, h_{k_1}), \text{Call}(\text{Op}_\ell, h_{k_2}), \text{Call}(\text{Op}_m, h_{k_3}), \text{Call}(\text{Op}_s, h_{k_4}))$, a Compose chain over the four detector operations.
- (iv) Each *unresolved channel* is a $\text{Hole}(x_k)$, a typed placeholder for a channel not yet fired.

The full program is the disjunction of all paths' Compose chains.

Theorem 1.3 (Compilation is Type-Preserving). *The compilation of a valid Tempus program P produces a vaHera fragment $\llbracket P \rrbracket$ that is well-typed under the partition-coordinate refinement type system (?), with each $\text{Call}(\text{Op}_s, \cdot)$ carrying a refinement predicate encoding the electric-dipole selection rules $\Delta\ell = \pm 1$, $\Delta m \in \{0, \pm 1\}$.*

Proof. Each detector operation Op_s is registered in the operation registry $\mathcal{R}^+$ (Definition 3.5 of ?) with: input refinement = the range constraint on the measured coordinate; output refinement = the conjunction of all selection rules applicable to that coordinate transition.

The Compositionality Lemma (?) ensures that the Compose chain of the four operations has a well-formed refinement type that is the conjunction of all per-stage refinements. Since each registered operation satisfies the selection-rule convention, the full Compose chain is well-typed iff every implied transition is admissible.

Any Tempus program that compiles syntactically from cells satisfying Equation (1) cannot imply a forbidden transition: the cell half-widths δ_i are measured timing residuals, which lie in the range $[0, \tau_{\text{BX}}/2]$ and correspond to S_t coordinates in $[0, 1]$; the angular and principal coordinates are extracted from physical detector hits. All such coordinates satisfy the partition bounds (??)–(??) of ?. Hence the compiled fragment is well-typed. $\square$

2 Kernel Instantiation for the LHC

The five-subsystem dispatch kernel $\mathcal{K} = \langle \text{Exec}, \text{PVE}, \text{CMM}, \text{PSS}, \text{DIC}, \text{TEM} \rangle$ (?) is instantiated as follows.

2.1 PVE: Partition Refinement Type Checker

The pre-dispatch validation engine is the typecheck function of the partition-coordinate refinement type system (?). It accepts a vaHera fragment iff:

- (i) all partition coordinate values lie within the bounds $n \geq 1$, $0 \leq \ell \leq n - 1$, $|m| \leq \ell$ (Theorem 3.1 of ?);

- (ii) every Compose-chain transition respects all registered selection rules (the Forbidden-Transition Theorem of ?);
- (iii) the three-route verification check on the event's partition invariant passes within the bound $2K \cdot 4^{-n_P}$ (?).

By the Decidability Theorem (?), this check runs in $\mathcal{O}(\|P\| \cdot |\star|)$ time, which for fixed trigger menu size and selection-rule registry is $\mathcal{O}(1)$ per event.

2.2 CMM: Timing-Interval Tree

The categorical memory manager is a balanced interval tree keyed on $\Delta P(k) \in (-\tau_{\text{BX}}/2, +\tau_{\text{BX}}/2)$. The lookup function retrieves the cached partition classification for any event whose timing residual vector $\{\Delta P(k)\}_{k \in \text{fired}}$ falls within an already-classified cell C_i . The insert function stores the classification keyed on the cell address.

Proposition 2.1 (CMM Lookup Complexity). *For a cell registry with $|\Gamma|$ cells and maximum depth d_Γ , the balanced interval tree provides $\mathcal{O}(\log |\Gamma|)$ lookup and $\mathcal{O}(\log |\Gamma|)$ insert.*

Proof. Standard balanced-tree analysis; the interval structure adds at most a constant factor per comparison. $\square$

2.3 PSS: Parallel Dispatcher for Commuting Subsystems

The pending-state scheduler implements parallel dispatch for all four detector subsystem operations:

Theorem 2.2 (Parallel PSS Dispatch). *Because the four detector subsystems (tracker/ $\hat{O}_n$, ECAL/ $\hat{O}_\ell$, HCAL/ $\hat{O}_m$, muon/ $\hat{O}_s$) have diagonal coupling $g^{(ij)} = 0$ for $i \neq j$ in the universal transport formula (?), the PSS may dispatch all four operations in parallel without contention. The parallel dispatch cost is $\max(T_{\text{tracker}}, T_{\text{ECAL}}, T_{\text{HCAL}}, T_{\text{muon}})$ rather than their sum.*

Proof. Diagonal coupling in the universal transport formula $X = \mathcal{N}^{-1} \sum_{i,j} \tau^{(ij)} g^{(ij)}$ is exactly the Jackson-independence condition (?): the throughput decomposes as $X = \mathcal{N}^{-1} \sum_i \tau^{(ii)} g^{(ii)}$ (per-class self-lags only, no cross terms).

In quantum language: diagonal coupling is equivalent to complete commutation $[\hat{O}_n, \hat{O}_\ell] = \dots = 0$ (Theorem ?? of ?). Commuting observables share a common eigenbasis and can be measured simultaneously without disturbance. In dispatch terms: the four operations share no internal state and have no data dependencies on each other. By the Per-Fragment Parallelism proposition (?), independent operations may be dispatched in parallel. $\square$

2.4 DIC: HLT Retrieval Controller

The retrieval subsystem controls what additional detector data the High Level Trigger requests from the

detector read-out. For a given L1-accepted event, the DIC implements the retrieval policy: for each HLT algorithm, fetch only the detector regions of interest (RoIs) identified by the L1 classification rather than the full detector readout. This is the “surgical retrieval” policy that reduces the HLT input bandwidth by the ratio (RoI area)/(full detector area) $\approx 1/300$ for a typical ATLAS event.

2.5 TEM: Composition Efficiency Monitor

The trigger event monitor accumulates: (a) the composition efficiency $\eta_C = 1 - \prod_{i \in I} (1 - \kappa_i)$ after each event, incrementing the numerator when at least one path fires (Definition of ?); (b) the universal transport coefficient $X = \mathcal{N}^{-1} \sum_{i,j} \tau^{(ij)} g^{(ij)}$ (Theorem 3.3 below); (c) the per-event partition uncertainty $\Delta \cdot \tau_{\text{BX}} \geq \hbar$ (verified per bunch crossing); (d) the per-channel timing residual distribution $\Delta P(k)$ for Kuramoto order parameter estimation.

All four quantities are emitted as dispatch events and consumed asynchronously, with no throttling of the dispatch path (the non-blocking TEM of ?).

3 Operational Theorems

3.1 Timing window from detector geometry

Theorem 3.1 (Timing Window from Universal Transport). *The LHC bunch crossing period $\tau_{\text{BX}} = 25$ ns is computed at kernel boot time from the ATLAS detector half-diameter $\Delta x = 7.5$ m and the speed of light via the universal transport formula:*

$$\tau_{\text{BX}} = \frac{\Delta x_{\text{detector}}}{c}, \quad (3)$$

where $c = \Delta x_{\text{ref}}/\tau_c$ is the speed of light derived from the partition transport primitive (?) with Δx_{ref} the reference path and τ_c the partition lag.

Proof. The universal transport primitive (?) gives $X = \mathcal{N}^{-1} \sum_{i,j} \tau^{(ij)} g^{(ij)}$. For the single-class limit ($\mathcal{N} = 1$, $g^{(11)} = 1$), this reduces to $X = \tau^{(11)}$ — the single-class mean lag. Applied to light propagation through the ATLAS detector volume with $\tau^{(11)} = \tau_c$ and the identification $c = \Delta x/\tau_c$ from the mass-transfer mechanism (?):

$$\tau_{\text{BX}} = X = \tau^{(11)} = \tau_c = \frac{\Delta x}{c} = \frac{7.5 \text{ m}}{3 \times 10^8 \text{ m/s}} = 25 \text{ ns}.$$

The kernel's reference frequency $f_{\text{ref}} = 1/\tau_{\text{BX}} = 40$ MHz is therefore a derived quantity from the deployment geometry, not a configured parameter. $\square$

Corollary 3.2 (HL-LHC Timing). *For the HL-LHC HGTD with $\sigma_t = 30$ ps timing resolution, the effective single-class lag is $\tau_c^{(\text{HGTD})} = 30$ ps, giving a fifth-coordinate resolution of $\delta S_t = 30 \text{ ps}/25 \text{ ns} = 1.2 \times 10^{-3}$ — one part in 833 of the bunch crossing period.*

3.2 Composition efficiency as transport coefficient

Theorem 3.3 (Composition Efficiency as Universal Transport). *The trigger menu’s composition efficiency η_C equals the normalised throughput of the universal transport formula:*

$$\eta_C = 1 - \prod_{i \in I} (1 - \kappa_i) = \frac{X}{\Sigma}, \quad (4)$$

where $X = \mathcal{N}^{-1} \sum_i \tau^{(ii)} \kappa_i$ is the trigger menu’s throughput coefficient with $\tau^{(ii)} = 1$ (normalised) and κ_i the individual path powers, and $\Sigma = 1$ is the ceiling.

Proof. The Compositionality Theorem of the universal transport primitive (?): for disjoint-support components (independent trigger paths with $C_k \cap C_{k'} = \emptyset$), $1 - X_{\text{agg}}/\Sigma = \prod_k (1 - X_k/\Sigma)$. The Multiplicative Cascade Law (?): $1 - \kappa_{\text{total}} = \prod_k (1 - \kappa_k)$. Identifying $X_k/\Sigma = \kappa_k$ gives $\eta_C = X_{\text{agg}}/\Sigma$, and with $\Sigma = 1$: $\eta_C = X_{\text{agg}}$. $\square$

Corollary 3.4 (Efficiency Ceiling from Saturation). *The TEM-monitored efficiency satisfies $\eta_C \leq \eta_C^* \approx 0.643$, the saturation ceiling of the catalyst cascade at $n_P = 60$ (Theorem ?? of ?). No trigger menu, however large, can exceed this ceiling without reducing τ_{BX} (moving to the HL-LHC timing).*

3.3 Three-route mass verification as PVE conformance test

Theorem 3.5 (Three-Route PVE Test). *The three routes to mass (accumulated residue, confinement cost, fixed-point of negation) from ? are implemented as three independently registered detector operations in the PVE’s operation registry $\mathcal{R}^+$. Their agreement to within $2K \cdot 4^{-n_P}$ at dispatch time is a $\mathcal{O}(1)$ conformance check embedded in the PVE.*

Proof. By the Computational Equivalence Theorem (?): three refinement-typed routes to the same invariant are all inhabitants of the same refinement type T_{n_P} after three-route verification. Implementing the three routes as three refined operations Op_{res} , Op_{conf} , Op_{neg} in $\mathcal{R}^+$, and registering the three-route check as the output refinement predicate on any operation that combines them, reduces the verification to a single typecheck call. By the Decidability Theorem (?), this check runs in $\mathcal{O}(3 \cdot |\star|) = \mathcal{O}(1)$ at fixed menu size. $\square$

3.4 Structural incorruptibility from the frozen surface

Theorem 3.6 (Structural Incorruptibility via Kernel Architecture). *A Tempus trigger program in which the PVE’s only accepted fragment type is $\text{Compose}(\text{Literal}(\Delta P(k_1)), \text{Literal}(\Delta P(k_2)), \dots)$ — Compose chains over timing-residual literals — has zero injection surface: no physical energy deposit can cause the PVE to accept a fragment that the timing residuals would not accept.*

Proof. The PVE’s function is a deterministic Boolean predicate on the syntactic structure of the vaHera fragment (?; Algorithm 1). A timing-only PVE accepts iff every node of the fragment is a Literal($\Delta P(k)$) for some channel k in the cell registry, or a Compose() over such literals. No other fragment variant is accepted.

A physical energy deposit in channel k changes $t_{\text{rec}}(k)$, hence $\Delta P(k) = T_{\text{ref}}(k) - t_{\text{rec}}(k)$. The PVE checks whether $|\Delta P(k)| < \delta_i$ for the relevant cell C_i . A high-energy deposit arriving at the wrong time increases $|\Delta P(k)|$; a deposit at the correct time is a genuine signal. In neither case does energy magnitude enter the PVE logic: the PVE accepts based on the timing residual value alone, not on the energy.

By the Independence Theorem (?), the PVE is functionally independent from the executor (which might measure energy). Energy deposits cannot propagate to the PVE’s interface method. The injection surface is empty. $\square$

3.5 HL-LHC upgrade stability

Theorem 3.7 (Stability Under Trigger Menu Upgrade). *Adding a fifth detector subsystem (HGTD timing, for HL-LHC) to the kernel by registering Op_π (parity coordinate) in $\mathcal{R}^+$ and adding the corresponding selection rules to $\star$ does not break any existing client of the frozen interface.*

Proof. By the Stability Contract Theorem (?): the kernel evolves additively — new operations are added to $\mathcal{R}^+$, new selection rules to $\star$, but no existing operation signature is changed or removed. Existing clients invoke only methods in the frozen surface of the pre-upgrade kernel, which are preserved unchanged. The upgrade is therefore backwards-compatible.

The Planck depth decreases from $n_P = 60$ to $n_P = 52$ (from ?), which tightens the three-route verification bound from 4^{-60} to 4^{-52} : the upgrade strictly improves verification precision without invalidating any existing conformance test (the tighter bound implies the looser bound). $\square$

4 The Tempus-Physics Bisimulation

4.1 Two operational semantics

Definition 4.1 (Physics LTS). The *physics labeled transition system* phys has configurations $(p_k, \text{trigger?})$ consisting of an event endpoint p_k in the partition hierarchy and a Boolean trigger flag. Transitions are backward navigation steps $p_j \rightarrow p_{j-1}$ via the parent map of ?. Terminal configurations are roots p_0 labelled with the physics process category (Higgs, W/Z , QCD, etc.). Observations: the L1 accept flag (true iff p_0 falls in the signal class).

Definition 4.2 (Kernel LTS). The *kernel labeled transition system* kernel has configurations $(\llbracket P \rrbracket, v)$ consisting

of a vaHera fragment and a partial result value. Transitions are kernel dispatch steps (Algorithm 1 of ?): validation, scheduling, CMM lookup, executor call, TEM observe. Terminal configurations are fully-evaluated fragments with value $v \in \{\text{accept}, \text{reject}, \perp\}$. Observations: v .

4.2 The bisimulation relation

Definition 4.3 (Bisimulation Relation). Define $R \subseteq_{\text{phys}} \times_{\text{kernel}}$ by: $(p_j, q) R (\llbracket P_j \rrbracket, v_j)$ if and only if:

- (i) p_j is at depth j in the partition hierarchy and $\llbracket P_j \rrbracket$ is the vaHera fragment representing the backward navigation state at step j ;
- (ii) the partition coordinates of p_j are encoded in the refinement types of the Compose chain at depth j ;
- (iii) $q = \text{true}$ iff the current path satisfies the trigger condition encoded in P .

4.3 The main bisimulation theorem

Theorem 4.4 (Tempus-Physics Bisimulation). *Let P be a valid Tempus trigger program with compiled fragment $\llbracket P \rrbracket$. Define the bijection $\phi :_{\text{phys}} \rightarrow_{\text{kernel}}$ by $\phi(p_k, q_0) = (\llbracket P \rrbracket, v_0)$ where p_k is the detected event endpoint and v_0 is the initial kernel state. Then R of Definition 4.3 is a bisimulation between $_{\text{phys}}$ and $_{\text{kernel}}$, and by the Soundness Theorem (?), they compute the same equation: for every input event, the physics trigger classification (backward completion to p_0) equals the kernel dispatch output (L1 accept/reject).*

Proof. We verify the four bisimulation clauses of Definition 3.1 of ?.

(B1) *Forward simulation.* Let $(p_j, q) R (\llbracket P_j \rrbracket, v_j)$ and suppose $_{\text{phys}}$ takes a backward step $p_j \rightarrow p_{j-1}$. We must find a kernel transition to a related configuration.

The backward step decrements the trit-address of p_j by one level (?). In the kernel, this corresponds to one Compose-chain evaluation step: the executor evaluates the (j) -th stage of the Compose chain, producing the partition coordinate at depth $j - 1$. The type system ensures the result satisfies all selection rules at that depth (by Theorem 1.3). The resulting configuration $(\llbracket P_{j-1} \rrbracket, v_{j-1})$ is related to (p_{j-1}, q) by the same construction as R .

(B2) *Backward simulation.* Symmetric: a kernel transition from $(\llbracket P_j \rrbracket, v_j)$ corresponds to one backward step in $_{\text{phys}}$, by reversing the above construction.

(B3) *Terminal observation, physics-to-kernel.* When p_0 is the terminal root configuration, the physics trigger fires iff p_0 is in the signal class. In the kernel, the fully-evaluated Compose chain produces $v_n \in \{\text{accept}, \text{reject}\}$ from the final stage.

By Dispatch Correctness (Theorem 4.1 of ?) and the Compilation Theorem (Theorem 1.3), the kernel's output on a well-typed event fragment equals what the bare executor (backward trajectory completion) would return. Since the bare executor IS the physics backward completion, the outputs agree.

(B4) *Symmetric to (B3).*

All four clauses hold; R is a bisimulation. By the Soundness Theorem (?), $_{\text{phys}}$ and $_{\text{kernel}}$ compute the same equation on every input event. $\square$

Corollary 4.5 (Determinism Coincidence). *Both $_{\text{phys}}$ and $_{\text{kernel}}$ are deterministic (each event has a unique backward completion trajectory, and the kernel's dispatch is a deterministic function of its inputs by ?). By the Determinism Coincidence Theorem (?), bisimulation reduces to extensional equivalence: same event in, same L1 flag out, for every event.*

5 Di-Muon Trigger: Complete Execution Trace

We trace a single $Z \rightarrow \mu^+ \mu^-$ event through every step of the runtime, grounding each step in a formal theorem.

5.1 Raw detector signals

A collision produces two muons of opposite sign with $p_{T,1} = 28 \text{ GeV}/c$ and $p_{T,2} = 22 \text{ GeV}/c$ in the ATLAS detector. The raw signals are:

- Inner tracker: two curved tracks with sagitta s_1, s_2 in the 2 T solenoid. Cyclotron frequency $\omega_{c,i} = qB/m_\mu$ (FT-ICR formula, Theorem ?? of ?).
- Muon spectrometer: two straight tracks in the toroid with curvature matching $p_{T,i}$.
- No ECAL deposits (muons are minimum-ionising).
- No HCAL deposits.
- Timing residuals: $\Delta P(k_1) = +0.3 \text{ ns}$, $\Delta P(k_2) = -0.4 \text{ ns}$ (both within cell half-width $\delta = 5 \text{ ns}$).

5.2 Step 1: Tempus compilation to vaHera

The di-muon Tempus program fires on two muon-spectrometer channels within timing cell $C_{\mu\mu}$:

$$C_{\mu\mu} = \{k : |\Delta P(k)| < 5 \text{ ns}, s_k = \pm \frac{1}{2}\},$$

$$P_{\mu\mu} = C_{\mu\mu}(k_1) \wedge C_{\mu\mu}(k_2) \wedge (s_{k_1} + s_{k_2} = 0).$$

The compilation $\llbracket P_{\mu\mu} \rrbracket$ produces:

$\text{Compose}(\text{Call}(\text{Op}_s, \text{Literal}(h_{k_1})), \text{Call}(\text{Op}_s, \text{Literal}(h_{k_2})), \text{Call}(\text{Op}_n,$

where Op_s extracts the chirality coordinate (charge sign from track curvature) and Op_n extracts the principal coordinate (momentum scale from cyclotron frequency).

5.3 Step 2: PVE type check

The PVE runs $\text{typecheck}(\llbracket P_{\mu\mu} \rrbracket, \Gamma, \star)$:

1. $\text{Literal}(h_{k_1})$: literal with $s_{k_1} = +\frac{1}{2}$ — in range, passes.
2. $\text{Call}(\text{Op}_s, h_{k_1})$: output refinement $s \in \{-\frac{1}{2}, +\frac{1}{2}\}$ with $\Delta s = 0$ (chirality preserved in muon propagation) — passes.
3. $\text{Call}(\text{Op}_s, h_{k_2})$: $s_{k_2} = -\frac{1}{2}$, passes.

4. Compose threading $s_{k_1} \rightarrow s_{k_2}$: $\Delta s = -1 \neq 0$? No — this is not a transition between the same particle's states; it is a pairing of two particles. The selection rules apply per-particle, and each particle's chirality is constant (muons don't change charge). Passes.
5. Principal coordinate check: $n_\mu = \lfloor \sqrt{p_T/p_{\text{ref}}} \rfloor + 1 = \lfloor \sqrt{25 \text{ GeV}} \rfloor + 1 = 6$ — in range for $n_P = 60$. Passes.
6. Three-route mass check: m_μ from cyclotron frequency (FT-ICR route), from muon spectrometer sagitta (TOF route), from calorimeter MIP hypothesis (Orbitrap route). Disagreement $< 2K \cdot 4^{-60}$. Passes.

PVE verdict: = true. Total: $\mathcal{O}(1)$ time.

5.4 Step 3: PSS parallel dispatch

The PSS schedules the four detector operations in parallel (Theorem 2.2):

- Op_n (tracker): measures $n_\mu = 6$ via cyclotron sagitta.
- Op_ℓ (ECAL): measures $\ell = 0$ (no shower = electromagnetic ground state).
- Op_m (HCAL): measures $m = 0$ (no hadronic deposit = azimuthal ground state).
- Op_s (muon): measures $s_1 = +\frac{1}{2}$, $s_2 = -\frac{1}{2}$.

All four fire simultaneously. Wall-clock time: $\max(T_{\text{tracker}}, T_{\text{ECAL}}, T_{\text{HCAL}}, T_{\text{muon}})$ rather than their sum.

5.5 Step 4: Backward trajectory completion

The executor performs backward completion from endpoint $p_k = (n = 6, \ell = 0, m = 0, s = \pm\frac{1}{2})$:

1. Step back: $\ell = 0, m = 0$ forces $\Delta\ell = 0$ at this level — the muon pair is in the $\ell = 0$ ground state of the Z decay.
2. Step back: $n = 6$ at the detector level corresponds to the Z boson partition level $n_Z = \lfloor \sqrt{m_Z/p_{\text{ref}}} \rfloor + 1 = \lfloor \sqrt{91 \text{ GeV}} \rfloor + 1 = 10$. The backward step navigates to $n = 10, \ell = 1$ (E1 transition at the Z vertex: $\Delta\ell = +1$ from the decay).
3. Root: $p_0 = Z \rightarrow \mu^+\mu^-$, category = electroweak signal.

By Theorem 4.4, this matches the physics backward completion of ? exactly.

5.6 Step 5: CMM lookup and insert

Before executor invocation, the CMM checks whether the timing-residual pattern $\{\Delta P(k_1) = +0.3 \text{ ns}, \Delta P(k_2) = -0.4 \text{ ns}\}$ has been classified before. On the first occurrence: miss, executor runs. On subsequent occurrences with the same cell address: hit, result returned directly. For di-muon events at the LHC rate ($\sim 10^4$ per second at design luminosity), most events are unique and the cache is cold; the CMM provides benefits for systematic signatures (recurring trigger seeds from beam halo, for example).

5.7 Step 6: L1 accept and TEM observation

The kernel returns $v = \text{accept}$ (the event falls in the signal category $Z \rightarrow \mu^+\mu^-$ with $p_T^\mu > 20 \text{ GeV}$). The TEM observes the dispatch event and increments: η_C numerator by 1 (one more accepted event); the X accumulator by the di-muon path's lag $\tau^{(\mu\mu, \mu\mu)}$; the ΔP histogram by the measured values.

Total kernel dispatch time for this event: $T_{\text{PVE}} + T_{\text{PSS}} + T_{\text{exec}} + T_{\text{TEM}} \approx 10 \text{ ns} + 0 + 200 \text{ ns} + 0 \approx 210 \text{ ns}$, well within the L1 latency budget of 2500 ns.

6 Discussion

6.1 What the bisimulation proves

Theorem 4.4 proves that the Tempus trigger program and the partition-physics backward completion compute the same equation on every input event. This is a stronger statement than “the trigger is efficient” or “the trigger is correct on test data”. It says the two systems are *observationally indistinguishable* — no event can produce a different classification in one system than the other.

The proof structure is constructive: we exhibit the bisimulation relation R explicitly (Definition 4.3) and verify the four clauses of Definition 3.1 of ?. This makes the proof machine-checkable in principle: the kernel's conformance test `validate_07_di_muon_pipeline` is the executable version of the bisimulation check for the di-muon case.

6.2 The universal transport formula unifies everything

The universal transport formula $X = \mathcal{N}^{-1} \sum_{i,j} \tau^{(ij)} g^{(ij)}$ (?) appears in three different roles in this paper:

1. As the formula for the timing window $\tau_{\text{BX}} = \tau^{(11)} = \Delta x/c$ (single-class specialisation).
2. As the formula for the composition efficiency $\eta_C = X/\Sigma$ (independent-class sum with diagonal g).
3. As the formula for the parallel dispatch correctness condition (diagonal g = Jackson independence = QND commutation).

The three applications use three different algebraic restrictions of g (rank-one, diagonal, diagonal), but all three derive from the same closed form. This is the formal reason the physics and the OS papers agree: they are both instances of the same universal structure.

6.3 Conformance as physics validation

Every conformance test in the kernel's test suite is simultaneously a physics validation. The ten LHC-specific tests listed in the collaboration plan (`validate_01` through `validate_10`) directly correspond to physics theorems of this monograph:

1. `validate_01_planck_depth` = Theorem 4.3 of ?.

2. validate_02_registry_immutability = Theorem 3.6.
3. validate_03_replay_rejection = Counting Irreversibility, Theorem 5.4 of ?.
4. validate_05_three_route_G = Three-Route Verification (?) applied to G .
5. validate_06_selection_rule_rejection = Forbidden-Transition Theorem (?).
6. validate_10_partition_uncertainty = Partition Uncertainty $\Delta \cdot \tau_{\text{BX}} \geq \hbar$ (?).

6.4 The loop is closed

The chain of derivations now runs from axiom to runtime without gaps:

Bounded Phase Space Axiom $\rightarrow$ Oscillatory Necessity $\rightarrow$
 Partition Coordinates $(n, \ell, m, s) \rightarrow C(n) = 2n^2 \rightarrow$
 Composition-Inflation $T(n, d) \rightarrow$ Planck Depth $n_P = 60 \rightarrow$ Four
 Analyzer Equations $\rightarrow$ Partition Lagrangian + Lorentz Force $\rightarrow$
 Ion = Partition Malformation $\rightarrow$ Three Routes to Mass $\rightarrow$
 $E = mc^2$ as Theorem $\rightarrow$ QND Commutation = Jackson
 Independence $\rightarrow$ Backward Trajectory $\mathcal{O}(\log_3 N) \rightarrow$ Selection
 Rules = Type Errors (vaHera) $\rightarrow$ Three-Route Verification (PVE)
 $\rightarrow$ Universal Transport (CMM/PSS/TEM) $\rightarrow$ Dispatch Kernel
 (Bisimulation) $\rightarrow$ Tempus Trigger Program $\rightarrow$ LHC L1 Decision.

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